Haotian Chen (Chahot)

MEng (by Research) Student

College of Computing and Data Science
Nanyang Technological University, Singapore

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Technical Blog (in Chinese)



Research Interests

Broad Areas — Quantum learning theory, quantum complexity, Fourier analysis of Boolean functions, and property testing, quantum information and cryptography.

Current Focus — Exact learning of k-Fourier-sparse Boolean functions from uniform quantum examples; testing and learning quantum juntas; upper and lower bounds in quantum query complexity.

Education

Aug. 2026 – Present — **Master of Engineering (by Research), Nanyang Technological University**, Singapore.
College of Computing and Data Science.
Supervisor: Prof. Pranjal Dutta.
Research area: Quantum query complexity, Fourier analysis, and testing and learning quantum juntas.
Tentative Thesis Topic: *Fourier-Analytic Methods for Quantum Learning and Testing of Structured Boolean Functions and Unitaries.*

Sep. 2021 – Nov. 2023 — **PhD in Computer Science and Engineering (Withdrawn), Seoul National University of Science and Technology**, Seoul, South Korea.
Department of Computer Science and Engineering.
Specialisation: Cybersecurity in IoT.
Research topic: *A Smart Contract and Digital Twin-based Heuristic Multi-Cooperation Scheduling Framework for Smart Manufacturing in IIoT Environments*.
GPA: 4.27/4.50 (97.4/100).
Programme discontinued following a change in research direction toward quantum computing.

Mar. 2017 – Aug. 2021 — **Bachelor of Science in Computer Science and Engineering, Seoul National University of Science and Technology**, Seoul, South Korea.
Department of Computer Science and Engineering.
Specialisation: Information Security.
Thesis: *Servers–Client Mini-Network Security Prototype Based on VMware and GNS3*.
GPA: 3.64/4.50 (90.2/100).
Rank: 10th among the department's Class of 2021 graduates.

Research & Professional Experience

Oct. 2024 – Apr. 2025 **National University of Singapore**, Singapore.
Research Assistant, Centre for Quantum Technologies.
Studied quantum authentication and capability certification in adversarial settings, including proof-of-quantumness-style protocols, unitary inversion, and trap-based verification techniques.

Dec. 2023 – Jul. 2024 **Publishing House of Electronics Industry**, Beijing, China.
Independent Author.
Authored a Chinese-language technical textbook on quantum security, including quantum key distribution, information-theoretic security, and post-quantum cryptography.

Sep. 2021 – Oct. 2023 **Seoul National University of Science and Technology**, Seoul, South Korea.
Ph.D. Scholar, Department of Computer Science and Engineering.
Undertook research training in cybersecurity, cryptographic systems, and distributed computing, including IoT security, blockchain, and digital twins.

Feb. 2022 – Nov. 2023 **Seoul National University of Science and Technology**, Seoul, South Korea.
Assistant Lab Manager, Ubiquitous Computing and Security Lab.
Assisted with grant proposals, project coordination, academic administration, and research collaboration.

Teaching & Academic Services

Mar. 2022 – Sep. 2023 **Teaching Assistant**, Seoul National University of Science and Technology.
Assisted with course administration, student coordination, examination design and logistics, grading, and academic records.

Sep. 2019 – Dec. 2020 **Physics Mentor**, Seoul National University of Science and Technology.
Provided one-on-one undergraduate tutoring in physics fundamentals and problem solving.

Skills

Languages Chinese (Native); English (IELTS 7.0); Korean (TOPIK Level 6).

Programming Python, Qiskit, Java, C++, LaTeX.

Software and Tools Docker, Wireshark, Nmap, Jupyter Notebooks, GNS3, VMware, OpenAI API, Scopus, Web of Science.

Honours & Professional Recognition

Scholarships

2017 – 2021 **Scholarship for Outstanding Achievement for Foreigners**, received each semester during undergraduate studies.

2021 – 2023 **Scholarship for Outstanding Achievement for Foreigners**, received each semester during graduate studies at SeoulTech.

Best Paper Awards

2022 World Congress on Information Technology Applications and Services, Jeju, South Korea.

International Conference on Big Data, IoT, and Cloud Computing, Jeju, South Korea.

International Conference on Future Information Technology, Applications and Services, Seoul, South Korea.

International Conference on Computer Science and its Applications, Vientiane, Laos.

Professional Activities

2023 **Alibaba Cloud Developer Community Blogging Expert**, Alibaba Changfengzhe Program.

CSDN Blogging Expert; more than 20,000 subscribers and approximately 600,000 visits.

Examiner, security section, Korean Information Processing Technology Professional Certification Examination.

Research Publications

Journal Articles

1. **H. Chen**, A. E. Azzaoui, H. Park, D. Camacho, and J. H. Park, "A Comprehensive Study of Quantum Computing Technologies in Smart City: Review and Future Directions," *Human-centric Computing and Information Sciences*, vol. 14, 2024.
2. S. R. Jeremiah, **H. Chen**, S. Gritzalis, and J. H. Park, "Leveraging Application Permissions and Network Traffic Attributes for Android Ransomware Detection," *Journal of Network and Computer Applications*, vol. 230, Art. no. 103950, 2024.
3. **H. Chen**, A. E. L. Azzaoui, S. R. Jeremiah, and J. H. Park, "A Novel Smart Contract Based Optimized Cloud Selection Framework for Efficient Multi-Party Computation," *Journal of Information Processing Systems*, vol. 19, no. 2, pp. 240–257, 2023.
4. **H. Chen**, S. R. Jeremiah, C. Lee, and J. H. Park, "A Digital Twin-Based Heuristic Multi-Cooperation Scheduling Framework for Smart Manufacturing in IIoT Environment," *Applied Sciences*, vol. 13, no. 3, Art. no. 1440, 2023.
5. A. E. L. Azzaoui, **H. Chen**, S. H. Kim, Y. Pan, and J. H. Park, "Blockchain-Based Distributed Information Hiding Framework for Data Privacy Preserving in Medical Supply Chain Systems," *Sensors*, vol. 22, no. 4, Art. no. 1371, 2022.

Conference Proceedings

1. **H. Chen** and J. H. Park, "Quantum-Powered Secure Multi-Party Collaboration Framework for Smart Manufacturing," in *Proceedings of the International Conference on Big Data, IoT, and Cloud Computing (BIC)*, Jeju, South Korea, Aug. 2023.
2. A. E. Azzaoui, **H. Chen**, and J. H. Park, "QAOA: A Quantum Approximate Optimization Algorithm for Optimal Base Station Selection in 6G Networks and Beyond," in *Proceedings of the International Conference on Computer Science and its Applications (CSA)*, Laos, Dec. 2022.
3. **H. Chen**, T. W. Kim, and J. H. Park, "Smart Contract-Based Secure Multi-Party Computation for Cloud Privacy Preserving," in *Proceedings of the World Congress on Information Technology Applications and Services (World IT Congress)*, Jeju, South Korea, Feb. 2022.
4. **H. Chen**, T. Kim, J. Park, and J. H. Park, "Optimized Node Selection Method for Efficient Cloud Computing Based on Smart Contracts," in *Proceedings of the Korean Information Processing Society Conference*, vol. 29, pp. 48–51, 2022.
5. **H. Chen** and J. H. Park, "BDT: Blockchain and Digital Twins-Based Secure Framework for Efficient Data Processing in Smart Manufacturing," in *Proceedings of the International Conference on Big Data, IoT, and Cloud Computing (BIC)*, Jeju, South Korea, Aug. 2022.
6. **H. Chen** and J. H. Park, "Quantum Homomorphic Cryptography-Based Efficient Medical Imaging Architecture for IoMT in Smart City Environments," in *Proceedings of the International Conference on Future Information Technology, Applications and Services (IFIT)*, Seoul, South Korea, Oct. 2022.

7 **H. Chen**, T. Kim, J. Park, and J. H. Park, "Secure Multi-Party Computation Based on Homomorphic Encryption for Privacy Preserving in IoT Networks," in *Proceedings of the Korea Information Processing Society Conference*, pp. 189–192, 2021.

8 **H. Chen**, T. Kim, and J. H. Park, "Multi-Blockchain System Based on Directed Acyclic Graph for Improving Data Processing Performance," in *Proceedings of the Korean Information Processing Society Conference*, vol. 28, pp. 25–28, 2021.

Book

1 **Chahot**, *Quantum Security: A New Era in Information Protection* (in Chinese). Beijing, China: Publishing House of Electronics Industry, 2024. ISBN: 978-7-121-48136-9. https://www.phei.com.cn/module/goods/wssd_content.jsp?bookid=65109.

References

Prof. Pranjal Dutta – Current Research Supervisor
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Nanyang Technological University, Singapore

Prof. Laurence T. Yang – Academic Mentor
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Zhengzhou University, China

Prof. Dongwann Kang – Course Lecturer
dongwann@seoultech.ac.kr
Department of Computer Science and Engineering
Seoul National University of Science and Technology, South Korea

Prof. Greg Thompson – Academic Instructor
gthompson@kaist.ac.kr
KAIST Language Center, South Korea